ASCIIgent: Agentic ASCII Art Generation

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https://github.com/HanaHasan04/asciigent

Abstract

Visual abstraction is the process of discarding redundant low-level visual data—textures, photometrics, and uninformative pixel noise—and re-encoding semantic structures into a reduced, symbolic visual language. While recent advances in neural sketch synthesis like SketchAgent demonstrate that vision-language models (VLMs) can synthesize continuous vector strokes, we investigate whether this spatial capacity extends to a fundamentally more constrained and discrete medium: **ASCII art**. In ASCII art, a model must convey recognizable silhouettes using printable characters on a fixed-size monospace character grid. Characters here serve a dual role: simultaneously functioning as language tokens and spatial micro-primitives. This dichotomy introduces a core question: do autoregressive models draw through spatial reasoning or pure textual memorization, and does active visual perception enable meaningful self-correction? We introduce **ASCIIgent**, a training-free framework evaluating monolithic prompting baselines, agentic discrete tool-use loops, and multimodal visual feedback across Gemini family models (`gemini-3-flash-preview` and `gemini-3.1-flash-lite`). Evaluating against structure-destroying scrambled controls to mitigate the text-reading bias of VLM judges, we reveal characteristic failure modes of monolithic generation (scanline drift, glyph collapse, and plan-truncation leakage) and show that iterative visual feedback enables grounded spatial refinement, localized editing, and frame-consistent animation.

1 Introduction and Related Work

Visual Abstraction and Agentic Sketching. Visual abstraction is a foundational capability of biological and computational perception [1]. Neural stroke models pioneered recurrent drawing representations [2], later advancing toward semantic vector sketching [3]. Recently, **SketchAgent** [4] formalized sketch generation as a continuous perception–reasoning–action loop: a frozen multimodal LLM iteratively emits Bézier stroke control points, inspects the updated rasterized canvas, and refines the drawing. Concurrently, **VideoSketcher** [5] demonstrated that dynamic sketching can be decomposed into an LLM-guided structural plan coupled with video diffusion priors. We adapt and generalize this agentic paradigm to discrete character media.

The ASCII Art Medium and its Challenges. ASCII art imposes a strict spatial constraint: characters must occupy fixed coordinates in a monospace grid $\Sigma^{H\times W}$. Unlike continuous splines, characters possess fixed discrete geometry: a forward slash `/` acts as a 45° line segment, an at-sign `@` acts as a dense tonal patch, and a period `.` functions as point stippling. Monolithic autoregressive generation suffers from **scanline drift**: generating left-to-right, top-to-bottom forces the model to commit to features in early rows that must align with geometry hundreds of tokens later without visual verification or backtracking. Public archives like `asciiart.eu` [6] document decades of human designs that language models have partially memorized during pretraining.

The VLM Evaluation Confound. Evaluating ASCII art with multimodal models introduces a severe confounding factor: as shown by Wang et al. [7] and ASCIIBench [8], VLMs systematically suffer from text bias, recognizing concepts by reading character sequences rather than perceiving geometric shapes. Our evaluation incorporates structure-destroying scrambled controls (*scramble-cells* and *scramble-ink*) to isolate genuine visual recognizability.

2 Methodology

2.1 Canvas Representation and Format Contract

We define the drawing canvas as a fixed character matrix $C \in \Sigma^{H\times W}$ with $H = 24$ rows, $W = 48$ columns, and printable ASCII alphabet $|\Sigma| = 95$ (codepoints 32–126). Deterministic monospace rasterization $\mathscr{R} : \Sigma^{24\times 48} \to \mathbb{R}^{H'\times W'\times 3}$ renders glyphs at exact cell offsets without font kerning. Every model response is validated by a strict parser: row/column overflows or missing code fences are repaired and logged in a `ParseReport`.

2.2 One-Shot Generation Baselines (P1)

We evaluate three one-shot prompting strategies across four target categories (`cat`, `dog`, `bear`, `horse`) and two Gemini family models (`gemini-3-flash-preview` and `gemini-3.1-flash-lite`):

1. **Zero-Shot:** Direct instruction: "Draw a {category} on the 24×48 canvas."
2. **Few-Shot:** Same prompt preceded by four hand-crafted exemplar categories (`fish`, `house`, `tree`, `sailboat`) strictly disjoint from test categories to prevent direct template copying.
3. **Procedural (CoT):** The model writes a structured 3-step spatial plan (major parts, row/column bounding regions, and contour glyph choices) before emitting the fenced 24×48 grid.

2.3 Agentic Loop and Turn Protocol (P2)

Instead of emitting all 24×48 characters monolithically, ASCIIgent interacts with an `AsciiCanvas` over a multi-turn loop with a budget of $T_{\max} = 3$ turns. In each turn $t \in \{1,2,3\}$, the agent emits a JSON batch of multiple discrete tool calls:

- `patch_edit`$(r,c,lines)$: Localized subgrid overwrite at 0-indexed (r,c) where alignment spaces preserve underlying ink.
- `draw_line`$(r_1,c_1,r_2,c_2,char)$ & `draw_rect`$(t,l,b,r,border,fill)$: Bresenham line and box primitives.
- `shade_density`(t,l,b,r,ℓ) & `erase_rect`(t,l,b,r): Tonal fill (' ' < '.' < ':' < '-' < '=' < '+' < '*' < '#' < '%' < '@') and clearing.
- `finish`: Early termination when the drawing is complete.

Why 3 Turns & What Happens Per Turn. A 3-turn budget mirrors a coarse-to-fine artistic workflow while bounding API latency: **Turn 1** drafts the coarse outer silhouette and primary facial landmarks from the text prompt; after Turn 1, the canvas is rasterized to a PNG bitmap and paired with a 2-digit row/column coordinate ruler; in **Turn 2**, the VLM inspects the rendered image, critiques missing anatomy, and issues `patch_edit` calls to add limbs and facial features; in **Turn 3**, the VLM performs a final visual verification to add fine secondary details (e.g., collar and tail) before calling `finish`. To isolate the role of vision, we ablate three feedback modes between turns: (1) **Visual (Ours):** rendered

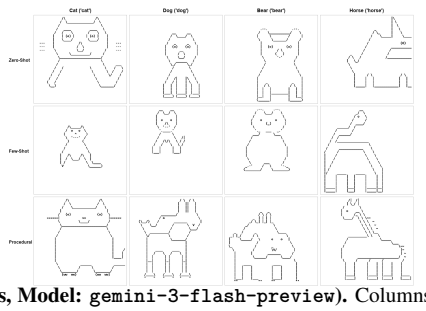


Figure 1: P1 One-Shot Baseline Contact Sheet (12 Cells, Model: `gemini-3-flash-preview`). Columns show the four target categories (`cat`, `dog`, `bear`, `horse`); rows show the three prompting arms. Zero-shot produces recognizable silhouettes but occasionally drifts across the 48-column right boundary (e.g., `horse`); few-shot regularizes scale and centering; procedural CoT (with bounded thinking effort `-effort low`) yields rich anatomical structure such as paws, manes, and hooves.

PNG bitmap + 2-digit coordinate grid; (2) **Text-Only:** raw character text without the rendered image; and (3) **No-Feedback:** blind multi-turn continuation without intermediate state.

2.4 Evaluation Protocol (P3)

Our evaluation combines three complementary pillars:

1. **Structural Metrics:** Ink ratio ρ (non-blank fraction), bounding-box coverage, Shannon character entropy $H = -\sum p_i \log_2 p_i$ (↑), distinct glyph count (↑), and automated defect flags: `spells_name` (↓) for lexical leakage and `degenerate` (↓). Note that ρ and coverage are structural descriptors without monotonic arrows because both extremes (0 blank vs. 1 solid block fill) are degenerate.
2. **VLM Image-vs-Text Judge:** Dual independent queries to a frozen multimodal judge (`gemini-3.1-flash-lite`): unprompted open identification ("What does this picture depict?") followed by a 1–5 visual quality rating (↑).
3. **Scrambled Controls & Control Gap (↑):** Every drawing is compared against *scramble-cells* (uniform permutation over the 24×48 grid) and *scramble-ink* (permutation of non-blank glyphs within the silhouette mask). The **Control Gap** $\Delta = S_{\text{real}} - \max(S_{\text{cells}}, S_{\text{ink}})$ confirms genuine geometric recognizability when $\Delta > 0$.

3 Experiments and Results

3.1 P1 Baselines and Multi-Model Comparison

Table 1 reports the complete per-cell metrics across all 12 cells for both `gemini-3-flash-preview` (Figure 1) and `gemini-3.1-flash-lite`, alongside the unbounded-reasoning ablation. Two key findings emerge:

- **Model Capacity vs. Glyph Collapse.** On the lightweight `gemini-3.1-flash-lite` model, zero-shot generation suffers from severe glyph collapse: `bear` zero-shot collapses into just 3 distinct characters (`|`, `:`, `.`) with entropy $H = 0.36$ bits and $\rho = 0.458$ (filling nearly half the canvas with solid vertical bars), while `cat` zero-shot uses only 4 distinct characters ($H = 1.74$ bits). In contrast, `gemini-3-flash-preview` synthesizes recognizable animal silhouettes across all 12 cells ($H = 2.33 \pm 0.32$ vs. 1.97 ± 0.57 bits; 8.33 vs. 7.33 distinct chars).
- **Reasoning Effort vs. CoT Truncation and Plan Leakage.** Because `gemini-3-flash-preview` bills internal thinking tokens against `max_tokens`, running the verbose `procedural` arm with unbounded reasoning (`effort=None`) caused **4/4 cells** to truncate: `cat` truncated after 2 lines (`degenerate=True`, $\rho = 0.005$), while `dog`, `bear`, and `horse` truncated mid-plan before opening a code fence (`no_code_fence`), leaking prose such as *"Step 1 – plan: a. The horse will be drawn..."* onto the canvas and triggering `spells_name=True` on **3/4 cells** (bottom row of Table 1). Bounding reasoning depth (`-effort low`) eliminated truncation across all 4/4 procedural cells (0/12 `spells_name`).

		`gemini-3-flash-preview`					`gemini-3.1-flash-lite`				
Cat.	**Arm**	ρ	Cov.	H↑	Ch↑	Sp↓	ρ	Cov.	H↑	Ch↑	Sp↓
Cat	zero-shot	.158	.78	2.28	8	No	.161	.78	1.74	4	No
Cat	few-shot	.055	.24	2.83	11	No	.094	.69	2.30	8	No
Cat	procedural	.138	.80	2.52	10	No	.102	.73	2.40	7	No
Dog	zero-shot	.102	.33	2.31	7	No	.070	.43	1.93	6	No
Dog	few-shot	.058	.20	2.58	8	No	.082	.43	2.42	8	No
Dog	procedural	.143	.68	2.15	9	No	.145	.71	2.18	9	No
Bear	zero-shot	.113	.46	2.56	10	No	.458	.71	0.36	3	No
Bear	few-shot	.082	.39	2.73	10	No	.148	.71	2.54	11	No
Bear	procedural	.128	.69	2.13	7	No	.230	.69	1.69	7	No
Horse	zero-shot	.098	.61	1.92	7	No	.194	.58	1.97	12	No
Horse	few-shot	.152	.67	1.76	5	No	.094	.38	1.97	5	No
Horse	procedural	.158	.76	2.21	8	No	.109	.63	2.08	8	No
Mean (n=12)		.115	.55	**2.33**	**8.33**	**0/12**	.157	.62	1.97	7.33	**0/12**
± SD (n=12)		±.037	±.22	±.32	±1.72	—	±.106	±.14	±.57	±2.64	—

Unbounded Thinking: `gemini-3-flash-preview` *procedural (*`effort=None`*,* n=4*):* 4/4 truncated; ρ=.331±.225, Cov.=.63±.41, H=4.05±1.73, Ch=38.5±23.9, Sp↓=**3/4**

Table 1: Complete P1 Structural Metrics Across Both Gemini Models (24 Clean Cells + 4 Truncated Ablation Calls). Arrows indicate directionality (H↑: Shannon entropy in bits, Ch↑: distinct chars, Sp↓: `spells_name` lexical leakage); ρ and Cov. are structural range descriptors.

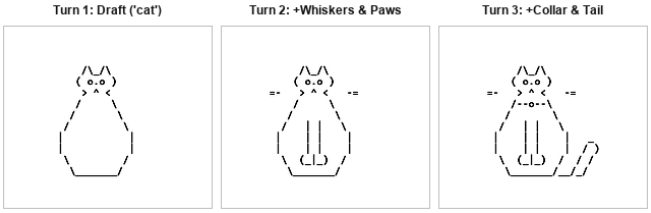


Figure 2: P2 Agentic Self-Perception Progression (Target: `cat`, Model: `gemini-3.1-flash-lite`). Turn 1 drafts the head and torso silhouette. Inspecting the rendered PNG in Turn 2 guides localized `patch_edit` calls for symmetric whiskers and seated front paws. Turn 3 adds a bell collar and curled tail.

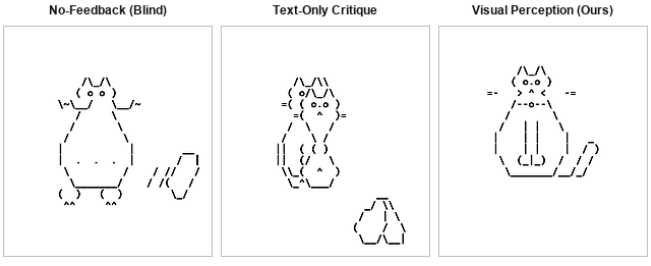


Figure 3: P2 Feedback Mode Ablation on `cat`. Left: `no_feedback` (blind) overwrites the nose and places floating disconnected feet/tail. Center: `text_only` critique misaligns row offsets, superimposing a second face over the first. Right: `visual` perception (ours) grounds every edit accurately.

3.2 P2: Agentic Synthesis and Feedback Ablation

Figure 2 illustrates the 3-turn agentic progression on `cat` using `gemini-3.1-flash-lite`. Even on the lightweight model, de-

composing generation into draft → render → perceive → revise yields a clean, high-entropy drawing ($H = 3.14$ bits, 13 distinct chars, $\rho = 0.056$, coverage $= 0.165$). As shown in the Figure 3 ablation, removing the rendered PNG causes severe spatial degradation: `no_feedback` ($\rho = 0.058, H = 2.89$) hallucinates disconnected limbs to the right of the body, while `text_only` ($\rho = 0.065, H = 2.92$) miscalculates row indices and stamps a duplicate face over the chest. Pairing the rendered PNG with the 2-digit coordinate ruler in `visual` mode eliminates both failure modes.

3.3 P3: CLIP Evaluation and Judge Calibration

Evaluating rendered drawings with local CLIP ViT-B/32 against text prompts (`"an ASCII art drawing of a {category}"`) confirms positive control gaps ($\Delta\uparrow$): the P2 `cat` achieves $S_{\text{real}} = 0.3251$ vs. 0.2737 (*scramble-cells*) and 0.2758 (*scramble-ink*), yielding $\Delta = +0.0493$; the P5 `bear` achieves $S_{\text{real}} = 0.2981$ vs. 0.2707 scrambled ($\Delta = +0.0274$); and P1 procedural `horse` (`effort=low`) achieves 0.3213 vs. 0.3088 ($\Delta = +0.0124$). Crucially, on the truncated `effort=None` procedural `horse` where leaked plan prose spelled `"horse"` (`spells_name=True`), CLIP yields $S_{\text{real}} = 0.1893$ vs. 0.2451 scrambled ($\Delta = -0.0558 < 0$), proving that Δ penalizes text leakage. In our dual VLM judge test (`run_judge_text_vs_image.py`), the rendered P2 `cat` bitmap is identified as `"cat"` and rated $5/5$ vs. $1/5$ scrambled ($\Delta_{\text{judge}} = +4.0$).

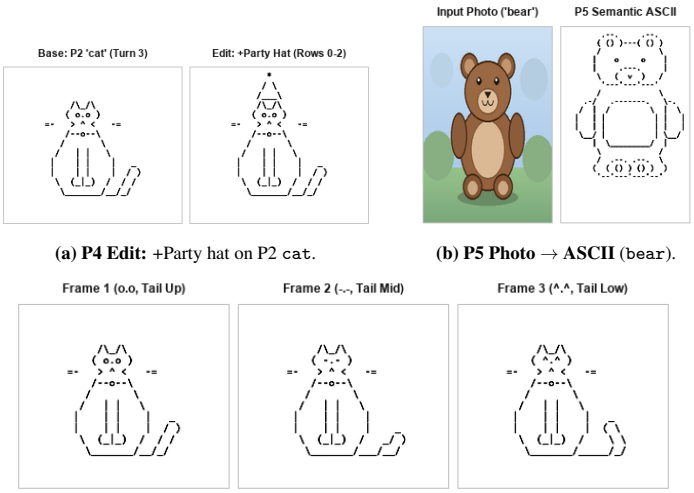
3.4 Unified Downstream Extensions (P4, P5, P6)

To ensure strict experimental continuity, both our conversational editing (P4) and keyframe animation (P6) experiments take the **exact Turn 3 `cat` canvas synthesized by our P2 agentic loop** (Figure 2, right) as their starting base drawing:

- **P4: Conversational Editing (Figure 4a).** Given the P2 `cat` canvas and the instruction *"Add a pointy party hat with a star on top of the head"*, the agent inspects the rendered image and 2-digit ruler and issues a single `patch_edit` at rows 0–2, columns 18–22 (modifying only 8 cells directly above the ears `/_/\` at row 3 while leaving rows 3–23 completely untouched; $\rho : 0.056 \rightarrow 0.0625$, 14 distinct chars).
- **P5: Semantic Image-to-ASCII (Figure 4b).** Given a 320×320 shaded RGB illustration of a seated `bear` on a sky-and-foliage background (`runs/p5_image2ascii/input_photo.png`), the VLM suppresses background clutter and photometric gradients to extract a clean 24×48 structural contour ($\rho = 0.125$, coverage $= 0.391$, $H = 3.04$ bits, 11 distinct chars).
- **P6: Delta-Patched Keyframe Animation (Figure 4c).** Regenerating a full 24×48 grid per frame causes body drift and limb mutation. Instead, `p6_animate.py` initializes each frame t from AsciiCanvas(base $=$ frame$_{t-1}$) and applies only localized `patch_edit` deltas. Starting from the P2 `cat`, the head, whiskers, collar, torso, and paws remain 100% invariant across frames while the eyes blink (`o.o` → `-.-` → `^.^`) and the tail wags.

4 Discussion and Conclusion

Comparison to SketchAgent and VideoSketcher. Whereas SketchAgent [4] emits continuous parametric Bézier strokes on a vector canvas, ASCIIgent operates over a *discrete character grid* via batched spatial primitives (`patch_edit`, `draw_line`, `shade_density`) paired with a 2-digit coordinate ruler. In VideoSketcher [5], an LLM decomposes a drawing plan into sequential strokes that a video diffusion model animates into pen-drawing footage. Because continuous diffusion does not apply to discrete text cells, ASCIIgent realizes dynamic synthesis in two discrete ways: (1) **procedural turn playback**, replaying the logged action sequence step-by-step (`progression.gif`), and (2) **delta-patched keyframe animation** (P6), where autoregressive visual conditioning and localized `patch_edit` operations enforce frame-to-frame body invariance.



(a) P4 Edit: +Party hat on P2 `cat`. **(b) P5 Photo → ASCII** (`bear`).

(c) P6 Delta-Patched Animation: Blinking eyes and wagging tail on the P2 `cat` with 100% invariant body and limbs.

Figure 4: Downstream Extensions (P4, P5, P6). P4 and P6 operate directly on the P2 Turn 3 `cat` canvas; P5 converts an RGB image of a `bear` into semantic ASCII contours.

Limitations and Future Work. Free-tier API rate limits constrain sample counts per cell, and VLM coordinate grounding still benefits from explicit 2-digit row/column rulers alongside rendered bitmaps. Future work includes expanding evaluations across additional open-weights and proprietary VLMs, calibrating the scrambled control gap Δ against human perceptual studies, and incorporating tree search over candidate discrete edits.

Conclusion. ASCIIgent demonstrates that while monolithic autoregressive generation is vulnerable to scanline drift, glyph collapse, and plan-truncation leakage, coupling a discrete spatial action vocabulary with active visual perception enables self-correcting ASCII synthesis, surgical conversational editing, semantic image-to-ASCII abstraction, and coherent keyframe animation.

References

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